

Presenting BIOXRE: project on BIOphilic design of Extended Reality working Environments

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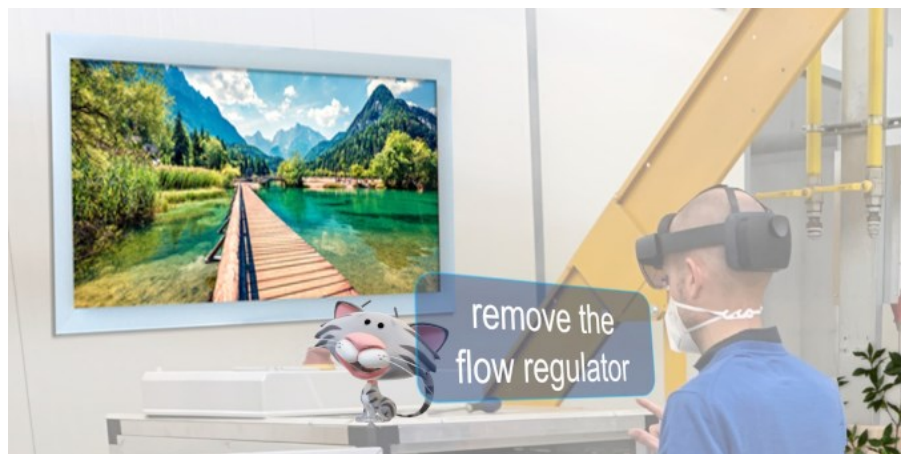


Figure 1: concept of the XR interface for the visualization of instructions with the biophilic design applied: besides procedural virtual instructions, BIOXRE proposes to enrich the XR environment (in this example, an Augmented Reality application for maintenance).

ABSTRACT

Nowadays, the technological maturity achieved by Extended Reality (XR) technologies allows to develop applications that foster positive emotions and promote personal growth. The scope of this work is to present the first results about a research project (BIOXRE - BIOphilic design of Extended Reality working Environments) focused on the application and evaluation of the biophilia hypothesis of Positive Computing in virtual workplaces. In the literature, there is effectively evidence that biophilic design improves human wellbeing. However, only a few studies evaluated its effectiveness in improving operators' wellbeing in the real workplace. For the first time, BIOXRE aims to exploit the biophilic design to XR applications supporting procedural tasks. The goal is to improve the wellbeing of human operators without affecting their performance in accomplishing manual tasks supported by XR visualization. The outcome of the project will be to extract guidelines for the spatial augmentation of real workplaces, fostered by the biophilic design, that will derive from a well-structured user evaluation in the target industrial scenario.

Index terms: Positive Computing, biophilic design, Industry 5.0.

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1 INTRODUCTION

Extended Reality (XR) is an umbrella term for any technology that alters reality by adding virtual elements to the real-world environment. It includes, but is not limited to, Augmented Reality (AR), Mixed Reality (MR) and Virtual Reality (VR). Over the last twenty years, numerous studies confirmed that XR is powerful to assist industrial operators, helping them in procedural tasks such as assembly and maintenance [1] or for training of novice operators [2]. Nowadays, the technological maturity achieved by XR technologies allows applying more advanced design paradigms beyond the basic users' needs and goals towards higher-level human needs. Consequently, it is possible to develop XR applications that foster positive emotions, promote personal growth and support creativity, thereby contributing to social advancement [3].

Sander has defined this approach as Positive Computing, that is "the study and development of information and communication technologies that are consciously designed to support people's psychological wellbeing in a way that honors the diverse ideas of the good life of individuals and communities." [4]. A relevant aspect in the application of Positive Computing is the improvement of working conditions starting with Indoor Environmental Quality (IEQ), which plays an essential role in affecting productivity, health, and wellbeing [5]. Even in the industrial field, which is traditionally focused on increasing the efficiency and flexibility of production [6], thanks to the novel paradigm of Industry 5.0, the wellbeing of industrial workers is placed at the center of the production process [7].

One of the approaches that can be used to improve IEQ consists of the biophilic design. It is based on the biophilia hypothesis that suggests humans have an innate connection to nature which may affect their health and productivity [8], [9]. The biophilic design incorporates natural structures (e.g., natural materials, façade

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greening, water) into the built environment to reconnect people with nature [10]. The application of the biophilic design to real working environments has been shown to positively impact an individual's wellbeing and reduce stress and pain [11]–[13].

Based on the results obtained in real working environments, we could hypothesize that the use of biophilic design to industrial XR applications could induce an improvement in terms of physiological stress and creativity as in a real scenario. Recently, the application of the biophilic hypothesis for improving wellbeing has also been studied concerning XR applications [14]–[16]. However, only a few studies evaluated the effectiveness of the biophilic design in improving operators' wellbeing in the workplace. In particular, [14] shows that people experiencing virtual work environments that mimic office spaces with biophilic elements had consistently lower levels of physiological stress indicators and higher creativity scores than those experiencing the same spaces without biophilic elements [14]. In the manufacturing field, to the best of the authors' knowledge, no study evaluated the application of the biophilic design to XR applications supporting procedural tasks.

In previous works, the biophilic design of XR applications consists of the introduction of biophilic virtual elements (BVEs) in the XR environment such as virtual plants and trees [17]. However, there are no design guidelines aiming at improving human wellbeing without reducing their performance in the accomplishment of the task using XR. They are important for the effective adoption of biophilic design in the development of XR industrial applications. The guidelines may regard, for example, the choice of the biophilic elements among different categories (e.g., plants, animals, landscapes), their placement in the XR environment, their resemblance to real biophilic elements, the exploitation of lights, colors, sounds.

With the aim of evaluating the human wellbeing as a consequence of the biophilia benefits, some studies have used different wearable sensors to examine the participants' physiological responses [14], [18], [19], however it is still unclear what are the best tools to obtain reliable data for this type of study. Recent works have used physiological responses, such as EEG, HRV, EDA, Blood Pressure, to recognize the emotional state of a subject in a XR environment by changing physical features of a room architecture in different fields [20]. Nevertheless, research quantifying physiological benefits of indoor biophilic features is sparse [21].

From the analysis of the state of the art, we can conclude that:

- There are only a few studies that propose the use of biophilic design in XR, in particular for industrial applications.
- The guidelines on how to apply the biophilic design in XR are missing.
- It is not clear how to evaluate human wellbeing to the exposure of biophilic elements.

Therefore, the overall objective of BIOXRE project is to propose and validate the use of BVEs in XR industrial applications with the scope of improving the wellbeing of human operators, without affecting their performance in the task accomplishment supported by XR visualization (Figure 1). Industrial use cases examples that can benefit from the results of the project are: the assembly of equipment supported by AR instructions, the training of operators in VR environments, the navigation inside large industrial plants exploiting AR. However, the findings of the project will also be useful for other jobs supported by XR, such as: virtual classrooms, offices in the metaverse, telemedicine. Throughout the duration of the BIOXRE project (two years, from 01/12/2023 to 30/11/2025), we will address all the issues related to the biophilic design of XR environments. In this paper, we report the first lessons learned and results achieved.

2 ACHIEVED RESULTS AND LESSONS LEARNED

2.1 Measurement of human wellbeing

The most traditionally used methods of well-being measurement consist of questionnaires and assessment scales. However, there is an increasingly prominent trend towards seeking objective data that can reflect well-being. Objective indicators are independent of subjective awareness, and objective evaluations rely on clear criteria and are measured impartially. In contrast, subjective measurements involve self-reporting based on implicit criteria, where individuals may misinterpret their situation due to several factors related to personal conditions and individual perceptions.

As previously mentioned, surveys are the most commonly used method of acquiring numerical data on people's well-being, allowing trends and patterns to be traced over time. In addition to surveys, techniques such as interviews and group discussions are used to gather qualitative information such as emotions and behavior over extended periods. These approaches integrate survey data by providing qualitative insights. Surveys are structured questionnaires designed to collect specific information on the well-being of individuals. They typically use closed-ended questions with pre-established response options, which facilitate the measurement and analysis of responses. These quantitative data are useful for monitoring trends and fluctuations in well-being over time, as numerical results can be easily compared and evaluated. Based on an exploration of diverse research studies and experiments, the most frequently utilized questionnaires are the following.

The Warwick-Edinburgh Mental Well-being Scale (WEMWBS) is a questionnaire consisting of 14 positive statements developed to assess the mental well-being of individuals.

The Positive and Negative Affect Schedule (PANAS) is one of the most used psychological questionnaires designed in 1988 to measure the two primary dimensions of mood or emotional states: positive affect (PA) and negative affect (NA).

The Satisfaction with Life Scale (SWLS) is a psychological evaluation instrument specifically designed to measure an individual's subjective well-being by assessing one's overall satisfaction with life.

The Perceived Stress Scale (PSS) is a frequently employed psychological instrument to measure the perception of stress in individuals' lives.

The State-Trait Anxiety Inventory (STAI) is a widely used assessment tool developed to measure both trait and state anxiety. It is extensively used for clinical purposes and in research settings.

The Self-Assessment Manikin (SAM) is a tool used to assess emotions and subjective feelings in response to a stimulus or situation.

The User Experience Questionnaire (UEQ) is a simple and standardized questionnaire used to rapidly evaluate the user experience of interactive products and systems.

In the field of social and personality psychology, the employment of self-report measures reviewed in the previous section has raised concerns due to potential user bias or conditioning.

Therefore, the need for more objective physiological and cognitive assessments to accurately measure impacts on well-being is recognized. Consequently, a growing inclination to study cognitive and physiological measures that can properly quantify well-being has emerged. Physiological measures of well-being encompass a range of indicators that reflect the physical and biological aspects of an individual's health and emotional state. These measures are used to assess various aspects of well-being, including stress levels, emotional regulation, and overall health. Through the use of electroencephalogram (EEG), galvanic skin



Figure 2: Biophilic design interface: selection of the plant category (a), guideline (b), and plant type (c).

conductance (GSC), skin temperature (ST), blood volume pulse (BVP), heart rate variability (HRV), and other measurement techniques, research has identified several variables associated with changes in brain and body activity that can detect fluctuations in emotions, highlighting a complex and non-linear relationship between emotions and physical states.

Physiological parameters commonly used to monitor human well-being in clinical settings, research, and virtual reality applications, which are of significant interest to us, are:

- Heart Rate Variability and Blood Pressure
- Electrodermal Activity
- Skin temperature
- Eye tracking
- Respiration
- Muscles activity
- Neuronal electrical activity

2.2 Development of a GUI for the placement of biophilic elements in the XR environment

For the placement of the BVEs in the XR environment for user testing, we implemented a simple but effective GUI (Figure 1) where the experimenter can set the following variables: 1) a selection of BVEs, from a database, that can be inserted in the XR environment; 2) the spacing among two BVEs; 3) the minimum and maximum distance from the user's eyes; 4) the quantity of BVEs (a fixed number or variable in a range); 5) the desired level of realism for the BVEs. The spacing and the distances, as well as the quantity of BVEs are defined following specific guidelines. Other variables may be introduced during the development of the project. Once the BVEs are placed in the Unity scene view, the experimenter can make little adjustments to their position as well as remove some elements for better tuning.



Figure 3: Examples of biophilic virtual elements from the database.

2.3 Database of biophilic elements

A meticulous search was conducted for models that could be incorporated within the project, accurately representing the plant types which have demonstrated positive effects, and which are typically integrated into work environments. They can be divided into four categories (Figure 3):

- Table Plants: *Schefflera arboricola*, *Marble Queen Pothos*, and *Chlorophytum comosum*, commonly known as spider plant.
- Floor Plants: included in this category are *Dracaena Trifasciata* (also known as snake plant or mother-in-law's tongue), *Ficus elastica*, and *Dracaena marginata*.
- Suspended Plants: species such as *Hoya linearis*, *Epipremnum aureum*, and *Philodendron scandens* Brazil are categorized here.
- Wall Plants: represented by a typical green wall.

3 CONCLUSION

In this work we present the first results achieved for the BIOXRE project. We made an overview of the available metrics for the measurement of human wellbeing as well as a selection of the type of plants that will be used for biophilic enrichment. This is the first step for future studies in which we will evaluate the effects on wellbeing of various biophilic designs. These different biophilic designs will be the result of the combination of various settings of biophilic properties (e.g., type of BVE, spatial distribution, quantity, level of realism) that can be easily made with the GUI that we developed.

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